

# ALESSANDRA GUERINONI

Senior Software Engineer & Neuroengineering Student

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## SUMMARY

Senior Software Engineer and Tech Lead at Tesla with 4 years' experience designing distributed systems (Kafka, Kubernetes, Go) and shipping AI-powered tooling at scale. Two promotions in 3 years; portfolio of \$22M+/year in validated business impact. Currently pursuing an MS in Biomedical Engineering at Johns Hopkins University with a focus in neuroengineering.

## EDUCATION

### Master of Science, Applied Biomedical Engineering

Sept 2025 - Present

Johns Hopkins University, Whiting School of Engineering – Engineering for Professionals

- Focus area: Neuroengineering.
- Completed coursework: Physiology for Applied BME I & II, Mathematical Methods for Applied BME.

### Bachelor of Engineering, Mechatronics Engineering

Sept 2016 - Apr 2022

McMaster University

- Awards: Golden Key Award (top 15% of the faculty); Future Leader Recognition Award; Dean's Honour List (3.7 GPA / 4.0 scale); Faculty of Engineering Entrance Award.

### Machine Learning Certificate

May 2024 - Aug 2024

Cornell University

## WORK EXPERIENCE

### Senior Software Engineer, Tech Lead

July 2022 - Present

Tesla

- Advanced from Associate to Senior Software Engineer in under 3 years (two promotions, Apr 2025).
- Led engineers building internal engineering platform tooling generating an estimated \$22M+/year in validated time savings across the organization.
- Designed and implemented a notification system using Kafka, Kubernetes, and Golang, generating an estimated \$18.8M in business impact.
- Architected and shipped a full-stack web application (Next.js, React, Go microservices, Kafka, S3) replacing legacy tooling — 10X performance improvement, 99.76% uptime, ~2000 daily users; reduced approval cycle time from ~20 days to 8 days (60% reduction); \$3.1M/year validated savings.
- Spearheaded AI integration into engineering workflows — production features achieving 97% classification accuracy, eliminating ~6,000 hours/year of manual review (\$1.2M/year validated savings).
- Engineered and optimized ETL pipelines for efficient data consumption via gRPC and REST APIs, leveraging Go, MySQL, and Temporal workflows.
- Mentored engineers and delegated major features while maintaining technical accountability; contributed to the hiring pipeline through technical screens and panel interviews; mentored intern through to full-time conversion.
- Assumed Acting Product Manager responsibilities (Apr 2026 - Present), owning backlog prioritization, product strategy, and stakeholder alignment after PM departure.

### Project Manager, Governance & University Collaboration

May 2019 - Aug 2020

Bombardier, Mitsubishi Heavy Industries

- Managed 50+ enterprise projects; led a team of 6 coordinating university partnerships and recruiting 26 interns; negotiated contracts reducing cost overhead by 2%.
- Built Python tooling for Oracle database access to streamline project reporting.

### Teaching Assistant – Discrete Mathematics & Computational Thinking

Jan 2019 - Apr 2019 & Sept 2021 - Dec 2021

McMaster Faculty of Engineering

- Facilitated Haskell and computer science learning for 50+ students; led tutorials, graded assignments, and ran consultations.

### Software Engineering Camp Counselor

Apr 2018 - Sept 2018

McMaster Faculty of Engineering

- Introduced 250+ children to Software Engineering through Python, Arduino, and web projects; delivered STEM lectures to 100+ students across Hamilton.

## PROJECTS

### TriAuto — Automated Emergency Triage System

Sept 2021 - Apr 2022

McMaster University Capstone

- Co-developed with a 4-person team a real-time decision-support system for emergency department triage — ingests patient intake data and produces a recommended acuity classification (CTAS 1–5), giving nurses a second-check during overwhelmed shifts when mis-triage risks delaying care. Stack: React, Node.js, MySQL, Socket.IO.

## SKILLS

### Languages:

Go • TypeScript • JavaScript • Python • Java • C++ • Haskell • MATLAB • SQL

### Cloud & Infrastructure:

AWS • Azure • Docker • Kubernetes • S3 • Kafka • Temporal • Redis

### Frameworks & Tools:

Next.js • React • Node.js • REST • gRPC • OAuth • JWT • Prometheus • Grafana • Splunk